

# **INSTITUTE OF ACTUARIES OF INDIA**

## **EXAMINATIONS**

**SA2 – Life Insurance**

**May 2025**

### **INDICATIVE SOLUTION**

#### **Introduction**

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

**Q.1)**

i) Following are the potential factors contributing to the observed decline in the company's EV growth rate compared to previous financial year:

**1. Value of New Business (VNB):**

The Value of New Business is the addition to the EV of the company due to writing of new business during the year. The reduction in VNB could be the reason for reduction in increase in EV of the Company. The reason of lower VNB could be as follows:

i) Lower new business sales:

A reduction in the rate of EV growth could be linked to slower new business sales. As the EV base grows larger, the company must write proportionally higher volumes of new business to sustain the same rate of growth in EV. If the company is maintaining or even reducing the volume of NB relative to the expanding EV base (assuming new business margins to be same), this could result in a visible reduction in the EV growth rate.

ii) Reduction in New Business Margin (NBM):

In addition to lower NB volumes, a decline in the new business margin (NBM) could contribute to the slower EV growth rate. Several factors could cause this decline, such as

- higher acquisition costs,
- re-pricing with lower margins,
- a shift in the business mix towards lower-margin products,
- a change in the profile mix towards lower-margin customers,
- unfavourable movement in the yield curves used for NBM reporting etc.

**2. Expected Return on EV:**

Another key driver of EV growth is the unwind rate, which is influenced by the expected return at the start of the year. The reduction in the expected return reduces the EV growth. Unless the returns have drastically fallen within a year, this factor may not be the sole contributing factor for such decline.

**3. Operating Variance:**

Variations in operating assumptions—such as expenses, mortality rates, and persistency—have influence on the growth of EV. If actual experience deviates negatively from these assumptions, the EV growth rate may decline. Such deviations might stem from initial assumptions that are not realized for example: unfavourable market conditions or increased mis selling affecting lapse rates, or catastrophic events such as pandemics impacting mortality experience. Additionally, an increase in the company's renewal expenses during the year could contribute to unfavourable experiences that hinder EV growth

**4. Changes in Operating assumptions:**

Unlike one-off events impacting EV as discussed above, a negative trend in operating experience—such as increased expenses, higher mortality rates, or reduced persistency—necessitates a reassessment and adjustment of assumptions for future projections. If actual experience consistently deviates unfavourably from current assumptions, it indicates that the existing assumptions need to be changed resulting in declined EV growth.

**5. Non-Operating Variance (Economic Variances):**

Non-operating variances primarily include changes in economic assumptions, such as equity returns or shifts in the RFR. For example, a change in the RFR curve at the valuation date leads to changes in assets and liabilities resulting in net impact on EV. This impact will vary between companies depending upon the ALM positions. Similarly, fall in equity market or other non-debt investments could have negative non-operating variances and thereby reduce the EV growth.

**5. Other Non-Operating Variance:**

Non-operating variances also include any ad hoc variances that fall outside of the scope of operating variances. The ad hoc variances could be any expense which is beyond management's control for example the expenses related to the ESOPs to the management are directly linked to the share price of the Company and hence could

be treated as other non-operating variance. The company's share price could have gone up over the year resulting in increased ESOPs related expenses and lower EV growth.

#### 6. Capital Movements and Dividend Payouts:

The movement in EV reflects capital activities, including equity infusions and dividend distributions. Capital raised through equity contributes directly to EV growth. In contrast, capital raise through debt exerts only an indirect influence on EV and does not fully impact its value. Similarly, dividend payouts to shareholders diminish EV growth by reducing the company's net assets. The company could have raised capital in previous year or could have paid higher dividends in current year, resulting in lower EV growth.

#### 7. Other Adjustments:

Adjustments such as corrections of estimation errors identified during the reporting period or modifications in methodologies—pertaining to components like the Present Value of Future Profits (PVFP), Time Value of Financial Options and Guarantees (TVFOG), Cost of Residual Non-Hedgeable Risks (CRNHR), or other previously unmodeled elements—can influence the EV growth rate. These changes may lead to either an increase or decrease in EV, depending on the nature and extent of the adjustments.

(Max 10)

### ii) Statutory Actuarial Surplus

- This represents the actual surplus in the revenue account, generated during the financial year, calculated in accordance with regulatory reserving requirements of the life insurance company.
- **Analysis of Surplus (AOS)** is the investigation of the sources of emergence of such surplus in revenue account during the year.

### Indian Embedded Value (IEV)

- EV comprises the expected future profits from the existing business (Value of In-Force, or VIF) plus the available net worth (NW). Changes in either VIF or NW influence the overall EV movement. Some changes may merely shift amounts between VIF and NW without affecting the overall EV. However, other factors can impact VIF and NW separately, thereby affecting the overall EV.
- **Analysis of Movement (AOM)** is the investigation of the sources of change in EV (separately for VIF and NW).

### Key Difference

Changes in NW could be considered as close proxy of changes in statutory actuarial surplus (barring components which impact NW but doesn't flow through the revenue account of the insurance company say dividend payouts, equity raise etc.) and hence for simplicity it could be assumed that AOS captures the sources of movement in NW whereas AOM captures the net impact of movement in both NW and VIF i.e. EV.

### 1. New Business Written During the Year

When a life insurance company writes new business, its EV typically increases, provided the new business written is profitable (i.e. has positive new business margins).

However, the impact of new business on the components of EV—VIF and NW—differs. While writing new business contributes positively to VIF by adding future profit streams, it can exert a negative strain on NW. This phenomenon of negative impact on NW, known as "new business strain," arises because insurers are required to set up statutory reserves at the inception of new policies. These reserves often exceed the (initial premiums less expenses/commission) received, leading to a temporary reduction in NW. However, in totality, the increase in VIF exceeds the reduction in NW by an amount equal to new business value created by selling the policy.

The statutory actuarial surplus does not capture the positive impact of increase in VIF but only the impact of decrease in NW. Hence, writing New Business is captured as positive item in the AOM whereas in AOS, it is captured as a negative impact on statutory surpluses.

### 2. Margins for Adverse Deviations Release

Throughout the year, life insurance companies experience a systematic shift in value from VIF to NW. This transfer occurs as profits embedded in VIF are realized and released, often through the release of MAD. As a

result, VIF decreases while NW correspondingly increases by an equal magnitude, leaving the overall EV unchanged.

Hence, MAD release is captured as no impact item in the AOM whereas in AOS, it is captured as a positive impact on statutory surpluses.

### 3. Operating Variances

Operating variances occur when the actual operating experience deviates from the assumptions used in the valuation, leading to impacts on the components of EV.

- **Mortality Variances:** A negative mortality variance—where actual mortality is worse than expected—leads to a decline in EV. Both VIF and NW are negatively impacted.
  - VIF declines because future profits are lost due to premature policy terminations from higher-than-expected deaths. This is a second order impact.
  - NW decline as the death benefits paid typically exceed the reserves held, leading to a strain on profitability.
  - Hence, negative mortality variance is captured as negative item in the AOM as well as in AOS.
- **Expense Variances:** A negative expense variance—where actual expenses exceed assumptions—also reduces EV.
  - In this case, VIF generally remains unaffected, as future expenses may not yet be adjusted.
  - However, NW decreases due to the higher-than-expected actual expenses incurred during the year.
  - Hence, negative expense variance is captured as negative item in the AOM as well as in AOS.
- **Persistency Variances:** The impact of a negative persistency variance—such as higher-than-expected lapses—can be more complex.
  - VIF may decline due to a loss of future profits from surrendered policies.
  - NW may increase, as surrender payouts are often lower than the reserves held, creating a release of surplus.
  - The net impact on EV depends on the relative magnitudes of these effects. Even though NW and statutory profits may improve in the short term, EV could still fall if the reduction in future profits (via VIF) outweighs the gain in NW.
  - Hence, persistency variance could be captured as positive or negative item in the AOM whereas in AOS, it is captured as a positive impact on statutory surpluses.

### 4. Non-Operating Variances (Economic Variances)

Since IEV is calculated on a market-consistent basis, changes in economic experience—such as shifts in the risk-free rate (RFR) curve or equity growth—affect EV growth. These economic variances primarily impact the VIF component, while the NW remains unaffected until unrealized gains or losses on the asset side are realized through selling. Upon realization, the profit or loss is recognized in the NW, accompanied by a corresponding reduction in VIF of the same amount.

Furthermore, changes in the RFR can influence the Valuation Rate of Interest (VRoI), affecting reserve estimations and leading to profit emergence or deferral. For instance, a reduction in RFR might lower the VRoI estimation, resulting in an increase in reserves. Consequently, the NW would decrease, with a corresponding increase in VIF by approximately the same amount.

### 5. Capital Movements / Dividend Payouts

Capital movements, such as equity capital raises or dividend payments to shareholders, directly influence the EV of a life insurance company. These transactions affect the NW component of EV but do not impact the VIF.

For instance, when a company raises equity capital, the influx increases the NW, thereby enhancing the EV. Conversely, dividend payments to shareholders reduce the NW, leading to a corresponding decrease in EV.

It's important to note that while these capital movements alter NW, they do not affect statutory profits. This is because such transactions impact the balance sheet or shareholder account directly, without influencing the revenue account.

Hence, capital movements are captured as positive or negative item in the AOM whereas in AOS, it is captured as a NIL impact on statutory surpluses.

### 6. Operating Assumption Changes

- Revisiting Best Estimate Assumptions:

If the credible experience suggests to adjust the best estimate assumptions, —such as higher expected mortality rates or reduced lapse rates—it typically leads to an increase in reserves. This increase reduces the NW, as more capital is allocated to meet future obligations. There could be second order impact on VIF decrease, the overall EV decreases due to the reduction in NW.

Hence, tightening best estimate operating assumptions is captured as negative item in the AOM as well as in AOS.

- Adjusting Margins for Adverse Deviations (MAD):

If the company increases the MAD without altering the underlying best estimate assumptions, the reserves increase, leading to a decrease in NW. However, since these additional margins are expected to be released as profits over time, the VIF correspondingly increases. This simultaneous decrease in NW and increase in VIF can result in NIL impact on the overall EV. Nonetheless, the higher required capital associated with increased reserves may elevate the Frictional Cost of Capital (FCoC), potentially reducing the overall EV marginally.

Hence, increasing MAD is captured as slight negative in the AOM whereas in AOS, it is captured as a negative impact on statutory surpluses.

(Max 16)

### iii)

**a)** The Chief Marketing Officer's (CMO) proposal to guarantee a minimum surrender value equal to 100% of premiums paid for non-participating (non-par) savings products is aimed at enhancing product competitiveness and customer appeal. It necessitates a thorough evaluation against established actuarial principles and assessments to ensure financial sustainability and equitable treatment of all policyholders.

#### Alignment with Actuarial Principles

- Equitable Treatment:** Ensuring fairness to policyholders at the time of surrender is a fundamental principle in life insurance. Mathematically, this equitable share is represented by the Notional Asset Share (NAS), which estimates the policy's value at the time of surrender. In the early years of a policy, the NAS is typically lower due to the recovery of high acquisition expenses (including commission) and risk charges. If a policyholder surrenders during this period and receives a surrender value exceeding the NAS, it may result in a disproportionate allocation of funds. This higher payout to early surrenders can impact the benefits available to persistent policyholders, potentially leading to higher premiums or reduced other benefits for them.
- Fair and Reasonable Surrender Values:** The surrender value payable must be fair and reasonable, considering the premiums paid, the duration of the policy, and the benefits accrued. The minimum surrender value equal to 100% of premiums paid may look fair and reasonable from the policyholder perspective given return of premium/capital. However, this may not be fair and reasonable from shareholders' perspective given it is not aligned with NAS.

However, it could also be considered fair for the policyholders to pay for the initial costs involved in their policy set up and underwriting etc, plus the risk cost for the period before surrender. Hence, some charge for these costs from the premium should be acceptable and appear fair & reasonable to the surrendering customers.

- Smooth Progression:** Surrender values should follow a smooth progression over the policy term and be close to the expected maturity value towards the end of the policy term. The current proposal demonstrates the smooth progression over the policy term with a minimum guaranteed surrender value.

(3)

#### b) Treating Customers Fairly (TCF) Considerations

- Enhanced Customer Value:** From a TCF perspective, offering a full refund of premiums upon surrender provides customers with greater flexibility and security, especially in unforeseen circumstances.

- **Potential for Mis-selling:** However, such guarantees might lead to aggressive sales tactics, emphasizing the surrender value over the policy's long-term benefits. This approach could result in customers purchasing policies that do not align with their financial goals, leading to dissatisfaction and potential reputational risks for the insurer including impacting persistency.
- **Market Differentiation:** While the proposal could position the company as customer-centric, it's essential to assess whether the increased surrender value would genuinely attract and retain customers or if it might lead to adverse selection. Also, whether the additional cost of higher surrender value is passed on to the persistent policyholders in form of higher premium or reduced benefits, or passed on to the shareholder in form of lower shareholder profits.

(2)

### c) Financial and Actuarial Implications

- **Profitability:** Offering minimum surrender value equal to 100% of premiums upon surrender, especially in the early policy years, can significantly impact the insurer's profitability. The initial years often involve higher expenses related to underwriting, commissions, and administrative costs, which are recovered over time. A 100% surrender value guarantee may necessitate higher premiums or reduced other benefits to maintain financial viability. If not, then this will impact the shareholder margins and put the financial viability of the product at risk.
- **Persistency** – Offering a surrender value equal to the total premiums paid can reduce policy persistency, especially when alternative investments yield higher returns amid rising interest rates. This may lead policyholders to surrender policies early,
- **Interest rate** – Non-participating life insurance policies primarily invest in fixed-income instruments like government bonds. When policyholders surrender early, especially under high surrender value norms, insurers may need to liquidate these assets to meet payouts. If interest rates have risen, bond prices fall, leading to potential losses upon sale. Without a Market Value Adjustment (MVA) mechanism, insurers bear this financial risk, as the surrender values may exceed the realizable asset values.

This might force the insurer to keep invested in short-term liquid assets like cash, t-bills, etc. which could compromise on the investment freedom and return from assets.

- **Product Design Flaw:** Even with adjusted benefits or premiums to maintain margins, if surrender values significantly exceed the policy's notional value, it indicates a fundamental design flaw. This issue is aggravated in the absence of a lock-in period, potentially leading to anti-selection, lapse re-entry, and tax avoidance concerns.
  - **Anti-Selection:** Guaranteeing high surrender values may attract individuals inclined to early termination, disrupting the risk pool and increasing the insurer's exposure to early claims and surrenders.
  - **Distributor Churning/ Lapse Re-entry Risk:** Without a lock-in period, distributors might exploit the system by initiating policies to earn commissions and then encouraging surrenders, allowing policyholders to reclaim premiums without affecting their principal, leading to potential mis-selling.

Even with a lock-in period, offering surrender values substantially higher than the net premiums (after deducting commissions) can incentivize agents and policyholders to collude, causing repeated lapses and re-entries, which are challenging to price accurately.

- **Tax implication:** There could be potential tax evasion in case premiums paid and benefits received are tax exempted. In such case, the policyholder may take benefit of tax on premiums paid but can get back the money tax free again in same year by surrendering the policy if surrenders are not taxed.
- **Capital and Reserving Requirements:** even if benefits are reduced/premiums increased to maintain margins, higher guaranteed surrender values increase the insurer's liabilities, potentially requiring additional capital to fund reserves and higher solvency requirements. This is because the reserves are floored to surrender value at policy level. This means reserves will increase materially for all such policies leading to lower statutory profits as well as higher capital requirements. Further, higher reserves will lead to higher Required solvency (RSM) as solvency capital is a factor based as percentage of reserves thus leading to further high capital requirements. This could affect the company's financial stability and its ability to invest in future growth.

(5)

iv) To ensure the financial viability of a non-participating savings product with early-year surrender values of 60%, 80%, and 100% of premiums paid in the first, second, and third policy years respectively, the company should implement the following checks and controls:

- Ensure that the proposed surrender values meet or exceed the minimum GSV as stipulated by the regulator. Compare the proposed surrender values with the Notional Asset Share (NAS), which reflects the policy's fair value based on premiums paid, expenses, and investment returns. If the surrender values exceed the NAS, the company may be exposed to persistency risks.
- **Cap Acquisition Costs:** Ensure that cumulative direct acquisition expenses (including commissions, incentives, and sales-related costs) do not exceed the difference between premiums paid and surrender value over the first three policy years. This measure aims to mitigate risks such as anti-selection, lapse re-entry, and scenarios where customers or distributors might profit more than their actual contributions.
- **Implement Claw-back Mechanisms:** If reducing commissions or incentives isn't feasible, introduce claw-back provisions tied to policy persistency. This approach allows insurers to recover a portion of the commissions if policies are surrendered prematurely.
- **Introduce Lock-In Periods:** Establish a lock-in period (e.g., five years) during which surrender values exceeding the minimum regulatory limits are inaccessible. During this period, policies that lapse due to non-payment would convert to paid-up status, with the enhanced surrender values becoming available only after the lock-in duration, discouraging early surrenders.
- **Apply Market Value Adjustments (MVA):** Incorporate MVA provisions for surrender values exceeding the regulatory minimums to adjust payouts based on prevailing market conditions, particularly interest rate fluctuations. This ensures that any losses from asset liquidation due to rising interest rates are appropriately accounted for, maintaining financial equilibrium.
- **Restrict Product Sales Volume:** Limit the distribution of such products to maintain overall financial viability, ensuring that the product's performance doesn't adversely impact the insurer's broader portfolio.
- **Selective Distribution Channels:** Offer these products through distribution channels with demonstrated high persistency rates and minimal instances of mis-selling, enhancing the quality and longevity of policyholder relationships.
- **Prerequisite Sales Thresholds:** Introduce these products only after achieving a predetermined sales volume of higher-margin products, ensuring that the insurer's profitability remains robust before venturing into offerings with potentially lower margins.
- **Adjust Margin Loadings:** Revise product pricing to incorporate sufficient margin loadings, accommodating higher surrender values and potential persistency uncertainties, thereby safeguarding the insurer's financial interests.
- **Perform stress tests** on persistency assumptions, particularly for the initial three years, to evaluate the impact of early surrenders on profitability metrics like NBM and shareholder IRR.

(Max 5)

[Max 41 Marks]

Q. 2)

i) **Audit Committee (mandatory)**

- Ensuring the integrity of financial statements and overseeing financial reporting processes.
- Establishing procedures to address concerns related to internal controls and checks.
- Overseeing the internal audit department and monitoring the rectification of identified irregularities.
- Recommending the appointment, remuneration, and performance evaluation of auditors, ensuring their independence.
- Acting as a compliance committee to monitor and report on significant compliance breaches.
- Approving or modifying related party transactions, ensuring compliance with applicable regulations.
- Approving additional work entrusted to auditors or their associates, maintaining the integrity of the audit relationship.

**Investment Committee (mandatory)**

- Recommending the investment policy and establishing the operational framework for investment operations, focusing on liquidity, compliance with prudential norms, risk management, and protection of policyholders' funds.
- Ensuring the effective implementation of the approved investment policy.
- Independently reviewing investment decisions, considering internal due diligence processes, and not solely relying on credit ratings for assessing credit and market risks.
- Formulating an effective reporting system to ensure compliance with the investment policy, including internal/concurrent audit mechanisms for ongoing monitoring.
- Meeting at least once a quarter to review investment operations and submitting reports to the Board on portfolio performance; approving Standard Operating Procedures for investment operations.

**Risk Management Committee (mandatory)**

- Establishing and recommending the Risk Management policy and framework to the Board.
- Setting risk tolerance limits and assessing the cost-benefit of risk exposures.
- Reviewing risk-reward performance to align with policy objectives.
- Advising on best practices in risk management and assisting the Board in effective operation of the risk management system.
- Maintaining an aggregated view of the insurer's risk profile across all categories.
- Advising on risk management decisions related to strategic and operational matters.
- Reporting to the Board on risk exposures and actions taken to manage them.

**Policyholder Protection, Grievance Redressal and Claims monitoring Committee (PPGR & CM Committee) (mandatory)**

- Implementing standard operating procedures to ensure fair treatment of customers, including setting time frames for policy and claims servicing.
- Creating effective systems to address complaints and grievances of policyholders, including those related to mis-selling by intermediaries.
- Setting up a framework to review awards given by Insurance Ombudsman and Consumer Forums, analyzing root causes of customer complaints, and advising management on rectifying systemic issues.
- Regularly reviewing the status of claims, including settled, rejected, and outstanding claims.
- Ensuring compliance with statutory requirements laid down in the regulatory framework and providing details of grievances at periodic intervals in prescribed formats.
- Ensuring that details of Insurance Ombudsman are provided to policyholders and that a Grievance Redressal Officer is in place, with their details available on the insurer's website.
- Reviewing claims reports, including the status of outstanding claims with aging, repudiated claims with analysis of reasons, and the settlement of unclaimed amounts on a quarterly basis.

**Nomination and Remuneration Committee (NRC) (mandatory)**

- Reviewing declarations of intending applicants before the appointment, reappointment, or election of directors by shareholders at General Meetings.
- Scrutinizing applications and details submitted by aspirants for appointment as KMPs, and making independent references to verify the accuracy of the information furnished.
- Ensuring that insurers obtain an annual declaration from Directors/KMPs confirming that the information provided at the time of appointment/reappointment has not undergone any change subsequently.
- Formulating the insurer's policy on remuneration packages and any compensation payments for the CEO, Executive and Non-Executive Directors, and KMPs, in alignment with the applicable framework.
- Ensuring that the remuneration packages of KMPs are in accordance with the Remuneration Policy approved by the Board.
- Ensuring that proposed appointments/reappointments of KMPs or Directors are in conformity with the Board-approved policy on retirement/superannuation.
- Being responsible for the succession planning of the insurer, including its implementation in a smooth manner.

**Corporate Social Responsibility Committee ('CSR Committee') (mandatory)**

- It is responsible for formulating and recommending the CSR Policy to the Board, outlining activities as specified in Schedule VII of the Act.
- The Committee recommends the amount of expenditure to be incurred on CSR activities and monitors the implementation of the CSR Policy.

- The CSR Policy, as approved by the Board, must be disclosed in the company's report and displayed on its website.

**With-Profits Committee (mandatory):**

- Ensuring compliance with the IRDAI's framework on with-profit fund management and asset share calculations.
- Overseeing the fair treatment of policyholders by monitoring surplus distribution and ensuring transparency in fund management.
- Reviewing and approving methodologies related to asset share calculations and surplus allocations.
- Providing recommendations to the Board on matters concerning with-profit policies and fund management.
- Ensuring that the management of with-profit funds aligns policyholder expectations and regulatory requirements.
- Submitting periodic reports to the Board and regulatory authorities as mandated.

**Ethics Committee (not mandatory)**

- Oversees adherence to external laws, regulations, and internal policies, including the company's code of ethics or conduct.
- Collects information on compliance activities, identifies weaknesses or breaches, and evaluates measures in place to address them.
- Manages confidential channels for reporting ethical concerns or violations, ensuring issues are addressed appropriately.
- Provides insights on ethical matters affecting the company's operations and assists in setting a culture of integrity from the top.
- Reviews and approves compliance initiatives, assesses their effectiveness, and addresses significant compliance issues.
- Monitors the implementation of the board-approved Code of Conduct, promoting transparency while maintaining necessary confidentiality.

(Max 10)

ii)

**a) Key features of group fund based non-linked insurance product as applicable under Indian Regulations:**

1. The group fund-based non-linked insurance products encompass the following categories:
  - a. Group Non-Linked Superannuation Products
  - b. Group Non-Linked Gratuity Products
  - c. Group Non-Linked Leave Encashment Products
  - d. Group Non-Linked Post-Retirement Medical Products
  - e. Group Non-Linked Products with significant savings elements
2. Insurers maintain separate accounts for all receipts and payments pertaining to each policy under group fund-based products. Interest rates are credited to each fund, along with any expenses charged.
3. The regulations don't prohibit insurer from giving such investment guarantees under Group fund based non-linked insurance products.
4. All group fund-based non-linked insurance products, except for superannuation products, are required to provide risk coverage, with explicit charges for mortality or morbidity risks. Superannuation products may offer risk coverage, provided there is an explicit premium charged for such coverage.
5. Group fund-based non-linked pension products can be structured as either Defined Benefit Schemes or Defined Contribution Schemes, offering flexibility in plan design.
6. These products can be offered under both employer-employee and non-employer-employee arrangements. In non-employer-employee homogeneous groups, the master policyholder typically acts as the trustee.
7. The minimum policy term for group fund-based products is one year, with provisions for annual renewals.

8. Policyholders can withdraw up to 25% of the fund annually without incurring a Market Value Adjustment (MVA). This feature provides liquidity for unforeseen payouts. Withdrawals exceeding the 25% annual limit may attract an MVA, which adjusts the fund's value based on prevailing market conditions. This mechanism helps manage the fund's stability and ensures fairness among all policyholders.

(Max 4)

**b) Purpose of such products from policyholder perspectives:**

1. The group fund-based non-linked insurance products could serve any of the following purpose of the policyholder:
  - To provide employees with a structured retirement savings plan, ensuring financial security post-retirement.
  - To manage and fund the statutory gratuity liabilities that arise when employees retire, resign, or pass away.
  - To manage the financial liability arising from accumulated leave balances that employees are entitled to encash upon retirement or resignation.
  - To provide medical benefits to employees after retirement, ensuring their health expenses are covered during their non-working years.
  - To offer employees savings-oriented benefits, such as bonuses or lump-sum payments, often linked to specific milestones or durations of service.
2. The product provides a fixed return on funds, determined at the policy's inception. This ensures predictability and aids in financial planning. Being non-linked means the returns are not tied to market performance, offering stability and reducing exposure to market volatility.

**3. Enhancing Employee Retention and Motivation**

Offering group insurance products enables employers to provide structured financial benefits, such as retirement savings and leave encashment. These benefits not only demonstrate a commitment to employee welfare but also serve as powerful tools for attracting and retaining talent. Employees are more likely to remain with organizations that invest in their long-term financial security.

**4. Professional Fund Management**

Insurance companies bring expertise in managing dedicated funds associated with these products. Their professional fund management ensures prudent investment strategies, adherence to regulatory norms, and optimal returns, thereby safeguarding the interests of both employers and employees.

**5. Facilitating Financial Planning**

By integrating these insurance products into their compensation structures, employers can systematically plan for future financial obligations, such as employee retirements or leave encashments. This proactive approach aids in budgeting and ensures that funds are available when needed, reducing unforeseen financial strains.

**6. Compliance with Accounting Standards**

Establishing dedicated funds through insurance arrangements assists employers in aligning with accounting standards related to employee benefit provisions. Such compliance not only enhances financial transparency but also ensures that organizations meet their statutory obligations efficiently.

(Max 4)

iii) From the insurer's perspective, introducing a guaranteed group fund-based non-linked insurance product with predefined returns and specific withdrawal provisions entails several risks. However, these can be effectively mitigated through strategic measures:

**1. Interest Rate Risk:**

- *Risk:* By guaranteeing returns over a 3 year period on single year renewable business, the insurer is exposed to changes in interest rates, resulting in reinvestments at lower rates or early sell of assets at depressed values.
- *Mitigation:* The insurer could assume certain level of renewability and payouts from the business. Basis that, investing in assets that closely match the duration and cash flow characteristics of the liabilities can help manage this risk. However, there could be experience variances from the assumptions made.

## 2. Withdrawal Risk

- *Risk:* Allowing policyholders to withdraw up to 25% of their funds annually without a Market Value Adjustment (MVA) can pose liquidity challenges, particularly if multiple policyholders opt for withdrawals simultaneously. Exceeding this threshold may necessitate the liquidation of assets, potentially at unfavourable market prices, thereby impacting the fund's overall value.

Due to the guarantee of asset returns at tranches level, complexities arise during withdrawals. It becomes challenging to accurately associate each withdrawal with its corresponding tranche, complicating the application of MVA. Additionally, determining the method for allocating guarantees to withdrawing policyholders is difficult, as identifying which policyholder belongs to which tranche is not straightforward.

- *Mitigation:*
  - Liquidity Management: Maintain a portion of the fund in highly liquid assets to meet withdrawal demands without disrupting the investment strategy. This approach ensures that the insurer can fulfill withdrawal requests promptly, even during periods of market volatility.
  - Utilization of Incoming Funds: Leverage new inflows in the fund to bolster liquidity requirements.
  - Transparent MVA Implementation: Clearly define MVA terms within the policy documentation, ensuring transparency with policyholders. The MVA serves as a protective mechanism, compensating for potential losses due to large, unanticipated withdrawals.
  - A clear policy must be established to determine the method—LIFO, FIFO, or insurer discretion—for asset liquidation. Additionally, a robust withdrawal process should be implemented to ensure timely and accurate benefit payments.

## 3. Renewability Risk

- *Risk:* The insurer faces the risk associated with the non-renewal of single-year renewable policies. While interest rates are guaranteed, policyholders retain the option not to renew, especially if more favourable rates are available from competitors.
- *Mitigation:*
  - Similar to Withdrawal Risk Mitigation: Since non-renewal can lead to fund outflows similar to withdrawals, the mitigation strategies are aligned. Leverage new inflows in the fund to bolster liquidity requirements.
  - Policy Design Considerations: Incorporate features that encourage policy renewal, such as loyalty additions at end of 3 year or increasing loyalty addition every year, to enhance retention rates and reduce the risk of non-renewal.

## 4. Operational Risk:

- *Risk:* Errors in policy administration, fund management, or communication can lead to financial losses or customer dissatisfaction.
- *Mitigation:* Implementing robust operational controls, regular audits, and staff training programs can minimize the likelihood of operational failures. Leveraging technology for automation can also enhance accuracy and efficiency.

## 5. Reputational Risk:

- *Risk:* Failure to meet guaranteed returns or manage withdrawals effectively can harm the insurer's reputation.
- *Mitigation:* Maintaining transparent communication with policyholders, promptly addressing concerns, and demonstrating a strong track record in fund management can bolster trust and reputation.

(Max 5)

iv) From a regulatory perspective, the objective of capital is to ensure that, in adversity, an insurer's obligations to policyholders will continue to be met as they fall due. Regulators should establish regulatory capital requirements at the level necessary to support this objective

The following are the rules and methods that should be specified by the regulator in the framework while developing the principle-based solvency assessment framework:

1. Establish clear instructions for valuing various asset classes to prevent inconsistencies among insurers.
2. Specify the cash flows to be considered in liability estimations and outline principles for setting assumptions used in projecting these cash flows.
3. Identify boundaries for asset and liability recognition, ensuring alignment with regulatory standards, if any.
4. Ensure consistent methods for recognizing and valuing assets and liabilities across all insurers when determining both available and required solvency capital, enhancing comparability across insurers.
5. Adopt a total balance sheet approach, considering all assets and liabilities, including those not directly backing insurance obligations.
6. The framework should identify the exhaustive list of risks that the insurance company is exposed to. Capture all material risks—underwriting, market, credit, operational, and liquidity—to ensure capital requirements align with each insurer's actual risk profile.
7. Allow for diversification benefits across different risks and lines of business, recognizing that not all risks materialize simultaneously. Calculate capital requirements for individual risks separately and then aggregate them, considering diversification benefits and correlations between risks.
8. Incorporate the effects of risk management techniques, such as reinsurance and hedging, into capital assessments. Reward insurers demonstrating strong risk management practices with potentially lower capital requirements, encouraging the enhancement of risk management systems.
9. Utilize relevant and credible data for calibrating risk stresses/factors, allowing insurers to use internal data where justified or a standard stress across all insurers.
10. Set a target confidence level (e.g., 99.5% over one year) to ensure sufficient capital to withstand adverse scenarios.
11. Define lower capital thresholds to trigger regulatory intervention, safeguarding policyholder interests.
12. Provide guidance on acceptable methods for capital assessment, including factor-based approaches and stress testing.
13. Provide guidance on calculating risk margins, representing compensation for the uncertainty in future experience. Where prescriptive methods are not mandated, offer insurers choices subject to regulatory approval.
14. Mandate clear and comprehensive disclosures about solvency positions, risk exposures, and capital adequacy to stakeholders, specifying the granularity of these disclosures, such as at the line of business level.
15. Data Verification: Mandate verification of data used in solvency assessments for accuracy and completeness
16. Implementation Strategy:
  - a. International Alignment: Ensure consistency with frameworks like Solvency II and IAIS standards.
  - b. Phased Implementation: Introduce the framework gradually to maintain market stability.

(Max 12)

v) The Financial Condition Report should normally include:

- The purpose of the report.
- The assessment of the financial position of the life office at the time the investigations were carried out.
- Any historical development of the life office which may have implications for the future of the life office.
- Comments on aspects of the historical development of the life office, such as any policy condition, prospectuses, methods of selling etc. which has led to Policyholders' Reasonable Expectations being formed with financial consequences for the life office.
- A review of the business of the life office and the environment in which it is operating.

- A review of the financial investigations with special reference to any future external factors which present a threat to the life office's financial security and the options open to the life office to deal with those factors.

The Appointed Actuary shall comment in greater detail on:

- The methods and assumptions that have been used. The report shall distinguish between those assumptions which are important and those which are less important as also those which are within the life office's control and those which depend on the external situation.
- Any changes made to the methods or assumptions used in the last similar report.
- Any inadequacies in his or her analysis due to factors such as non-availability of data or any other matter.
- The differing nature of the assumptions and the extent to which those can be relied upon. He or she shall also comment on the factors that are outside the control of the life office.
- How the assumptions made compare with the recent experience and explain the reasons where the assumptions significantly differ from the recent experience.
- Any specific financial characteristics of the existing business or new business being written that may require special consideration.

The report should indicate, where appropriate, how the following items have been addressed:

- Any unusual features of the existing business, new business being written or proposed product development.
- Any financially significant reinsurance arrangement.
- Effects of any known or planned changes in the life office's method of operation.
- Any use made by the life office of derivatives, its purpose and its financial significance.
- Post Balance Sheet events/ developments since the end of the financial year/report period.

The Appointed Actuary needs to be vigilant about the factors that might affect the life office, such as:

- Concentration of assets in particular risk areas;
- Derivatives;
- Assets containing unusual provisions which may be susceptible to particular risks;
- Sources of new business which have unusual characteristics;
- Impending major claims or litigation that might affect the life office;
- Risks arising out of product literature or policy documentation;
- Loss of a distribution channel;
- The impact of options and guarantees in the insurance liabilities in different scenarios.

If the Appointed Actuary identifies situations or sets of situations, which lead or could lead to financial difficulty for which no satisfactory remedial action can be found, the Appointed Actuary should ensure that such matters have special prominence in the report and include any recommendations as to what might be done to avoid such circumstances.

(Max 10)

**vi) The Interest Rate Risk Framework should capture the following:**

- Identification
- Measurement
- Management and
- Monitoring of interest rate-related risks

**1. Identification of the Risk**

- Interest rate risk encompasses the potential adverse effects of interest rate fluctuations on a company's financial health.
- **Valuation Impact:** Changes in interest rates can alter the present value of assets and liabilities, thereby affecting key financial metrics like embedded value and solvency ratio on statutory basis.
- **Cash Flow Management:** Interest rate movements influence the timing and amount of cash flows from assets and liabilities. Effective ALM strategies are essential to mitigate reinvestment risk and ensure liquidity.

## 2. Measurement of the risk:

This section highlights the Key Risk Indicators related to interest rate risk that need to be developed for measuring the risk in the business. Below is the list of Key Risk Indicators that could be considered:

- **Cash Flow Matching:** Structuring asset cash flows to align with liability outflows, minimizing reinvestment risk.
- **Duration Matching:** Aligning asset and liability durations to reduce exposure to interest rate movements.
- **Dollar Duration Gap:** Measuring the difference in duration between assets and liabilities in absolute monetary value, indicating how changes in interest rates affect the economic surplus.
- **Key Rate Duration:** While dollar duration captures the impact of parallel shifts in the yield curve, key rate duration measures the sensitivity to non-parallel shifts at different points along the curve.
- **Economic Capital Requirement for Interest Rate Risk:** Estimating the capital needed to cover potential losses under various interest rate scenarios.
- **Liquidity Ratio:** Assessing the surplus of liquid assets and operating cash flows over expected cash outflows to ensure adequate liquidity.
- **Solvency Ratio Sensitivity:** Evaluating the impact of interest rate stresses on the solvency position.
- **Guarantee Rates Monitoring / Breakeven rate monitoring:** Tracking the minimum investment returns required across different business lines to meet all guaranteed liabilities.
- **Policyholder Behavior Modeling:** Monitoring and modeling actions such as surrenders and withdrawals, which may vary with changes in interest rates.

## 3. Management of the risks:

The interest rate risks should be managed using a range of policies to ensure that the investment activities and asset positions are appropriate to company's liability and risk profiles. Below are some of the policies to be considered:

- **Controls in Investment Strategy or Hedging:**
  - **Utilization of Derivative Instruments:** Employ interest rate swaps, options, futures, and other derivatives to hedge against unfavorable interest rate movements, thereby stabilizing cash flows and preserving asset values.
  - **Determination of Hedging Extent:** Conduct thorough analyses to decide on the appropriate level of hedging, considering factors such as cost, risk tolerance, market outlook, and the specific characteristics of the liabilities and assets involved.
  - **Portfolio-Level Offsetting:** Leverage offsetting positions within the broader investment portfolio to mitigate net interest rate exposure, optimizing the overall risk-return profile.
  - **Hedging Strategy Performance Evaluation:** Regularly assess the effectiveness of implemented hedging strategies to ensure they meet desired risk mitigation outcomes and adjust tactics as necessary.
  - **Dynamic Portfolio Rebalancing:** Continuously adjust the investment portfolio to reflect changes in interest rates and evolving market conditions, ensuring alignment with the organization's risk appetite and return objectives.
- **Setting risk limits or tolerance limits on key performance indicators like**
  - Sensitivity of Solvency ratio
  - Liquidity ratio
  - Economic Capital due to Interest rate risk
  - EV sensitivity
  - Dollar Duration / Key rate duration gap

- Strategic Asset Allocations to assess the risk-reward trade off, appropriate risk return metrics should be identified.

#### 4. Monitoring

Monitoring and reporting of the risk on ongoing basis through development of control and reporting procedures. The framework should also talk about the governance structure.

- Provide the Board and senior management with timely reports on interest rate risk exposures and management actions.
- Ensure adherence to regulatory requirements related to interest rate risk management and reporting.
- Specify the frequency and conditions under which the framework will get reviewed.

(Max 14)

**[59 Marks]**

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